Assessed Coursework Cover Sheet for PSYC0101

Complete and insert a copy of this cover sheet as the first page of every piece of assessed coursework you submit for PSYC0101.

STUDENT CANDIDATE NUMBER (given by Examinations and available on Portico): YPVH2
MODULE NAME & CODE: PSYC0101: INTERACTION SCIENCE
COURSEWORK PART (IF RELEVANT): PART 2
NAME OF TEACHER WHO SET THIS WORK: ANEESHA SINGH & DUNCAN BRUMBY.
COURSEWORK TITLE: HCI RESEARCH METHODS
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Coursework 2 – Part A: Qualitative HCI Research Methods

Section A1: Writing up the study

Aim

Sensemaking focuses on explaining why people seek and how people utilise information. Specifically, it describes the iterative interaction of information-seeking, meaning ascription and action (Thomas et al., 1993). Students enrolling in UCL's MSc Programmes face the challenge of making sense of and taking action upon various information under different scenarios. The purpose of this study is to understand how sensemaking theories apply to this situation. A series of semi-structured interviews designed to understand how prospective students gathered and used information to decide joining an MSc programme in the year 2022 was qualitatively analysed to achieve this purpose.

Background

Established studies (Pirolli & Card, 2005; Klein et al., 2007; Blandford & Attfield, 2010) on sensemaking commonly describe it as an iterative loop that generates internal representations of information and seeks for new information guided by existing representations. Pirolli and Card (2005) constructed the Sensemaking model that spotlights two bi-directional iterative loops: the information foraging loop and the sense-making loop. Holistically, this model can be considered in both the bottom-up (obtained data informing theory) and top-down (theory informing the search of data) approach. A core concepts introduced in this model, schema, encapsulates the cognitive process of selectively generating patterns around important task elements during an extensive experience. Klein et al. (2007) proposed the Data/Frame model that shares similarities with schema. It is again an iterative model where data constructs the frame, and the frame determines data selection. Blandford and Attfield (2010) advanced the interpretation of the sensemaking further by developing the adaptive Information Journey model emphasising the evolution of the understanding of and interaction with information, and the serendipitous nature of the sensemaking mechanism.

In the case of students joining a master's programme, the complex reality makes their decision-making process challenging. This study will explore how the three models apply to this challenging process.

Method

6 participants were recruited from UCL's MSc Human-computer Interaction cohort by a group of 3 researchers in the same course, each took part in a one-on-one semi-structured interview around the topic "Why did you choose to study MSc HCI at UCL?" The interviews focused on how internal representations guided the foraging and interpretation of new information and were reciprocally informed by information.

During the initial construction of the interview plan, the PRETA framework (Blandford et al., 2016) was referred to as guidance to ensure its effectiveness. To communicate the purpose of the study and participants' rights, consent was obtained by them signing a physical consent form and verbally confirming immediately before the interview.

During the interview, one researcher actively communicated with the participant loosely following an interview plan, and another observed, took notes and recorded the audio. After each round, suggestions for improvement (if any) were made by the observer to the interviewer. After 3 rounds of interviews, the interview plan was reviewed and refined. Subsequently, another 3 interviews were conducted on a separate day. Audio recordings and hand-typed transcriptions were stored locally in compliance with GDPR. Transcripts were anonymised.

A two-way thematic analysis (Blandford et al., 2016) on the 6 transcripts was conducted by the group. First, a preliminary list of codes based on information-seeking and sensemaking theories was selectively assigned to the transcripts (top-down method), and then open codes were generated (bottom-up method). Finally, themes were generated by analysing and organising all the codes.

Results

Among several themes obtained through the thematic analysis, two were particularly germane to the research topic:

Theme 1: Information Aggregation for interpretation

When deciding whether to apply for a certain course, 5 out of 6 participants gathered information extensively from multiple sources before generating any internal representation. Then an educated interpretation of the holistic situation informed by these pieces of information was generated, to determine what the further action should be.

For example, one participant stated that they went to the UCL website to "look at the course structure […] lectures, the index of […] their publications, and how much they've published" to determine whether they are going to apply. Another mentioned that before making the decision to apply "it was mostly Googling and doing research, looking at their websites, and then attending some open campus tours […] (and) university open days."

Theme 2: Action Informed by Objective Data

When facing the urgency of making an important decision, quantitative data tends to be quickly interpreted and used to inform a physical action, whereas qualitative information is likely to be validated and interpreted more extensively and tends to inform the need for more information by creating uneasy emotions.

For example, when deciding whether to accept the offer from UCL, one participant "compared their university ranking and decided to go to UCL" determinately. Another two participants became anxious over qualitative factors, as one mentioned "Reddit didn't really

help because everyone was saying there's no point in doing HCI" and the other stated there was a lack of clues to "imagine what life will be" after joining the program.

Discussion

Theme 1 reveals a sensemaking process where multiple pieces of information were selectively obtained to form a personalized understanding of the environment, which will be used to direct further action. This process highly aligns with the concept of schema generation in Pirolli and Card's (2005) Sensemaking model, in a bottom-up manner. It also supports the Data/Frame model proposed by Klein et al. (2007) as it shares similarities with the schema concept

Theme 2 supports the serendipitous attribute of Blandford and Attfield's (2010) model. It shows that the flow of information journey could depend on both external (information characteristic) and internal (emotion) factors.

However, both themes have challenged the existing sensemaking models as they pressed the importance of emotional factors, and by extension, the context that impacts emotion. Out of the three, the Information Journey Model (Blandford & Attfield, 2010) best adapts to this phenomenon due to its flexibility in accounting for the information validation and interpretation process, hence has the most potential to be further developed to better define this phenomenon.

Due to the small sample size, and the common interest between researcher and participants, the generalisability of the result is limited. A future study conducted by an independent research entity on a larger and more diverse sample would refine the findings.

Conclusion

This study has validated the Sensemaking model (Pirolli and Card, 2005) and Data/Frame model (Klein et al., 2007) by revealing how internal representations are generated and treated. It has also suggested the potential of extending the Information Journey model (Blandford & Attfield, 2010) by pointing out the impact of information characteristics and emotional factors. These findings encourage further research to extend the theories of sensemaking.

Word count: 1096/1000

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Section A2: Writing up rationale & lessons learned

Before conducting semi-structured interviews, I familiarised myself with multiple sensemaking theories to come up with an interview plan. A considerable amount of effort was spent on contemplating the possible responses from participants and how my questions should follow through. I also focused on making the questions sound friendly.

During the first interview, the opening questions the ground for more cognitively demanding questions by making the participant comfortable and enthusiastic. This is because the opening questions were framed in a conversational way, and the tone showed my curiosity, which encouraged enthusiastic responses. For example, after being asked “Do you mind telling me where you live right now?”, “Do you like it?”, “Where were you last year?” and “How was that?” the participant would voluntarily compare their past living experience with the current. When questions about joining the course were asked, they were ready to share more about their complex experiences.

However, in the middle of the interview when questions became specific, the conversation started to deviate from research aim. This was because when the participant was stimulated to talk about their specific experience, I could not find an appropriate response from the plan, and eventually asked a less meaningful follow-up question. For example, when the participant said, “I was super attracted by the course content and now I do find myself enjoying it.”, in order to make the conversation natural while struggling to direct it back on track, I responded with “Which course do you enjoy the most and why?”, resulted in a long and invaluable answer.

When observing my teammate conducting their interviews, similar patterns were spotted. I noticed that at the beginning, effective questions were all simple yet inspiring, which comforts the participant. In the middle, as the conversation progressed, spontaneous and creative questions worked better than planned questions, as they will keep the response relevant and contextual. In the end, specific yet shallow questions are appreciated as they keep participants calm and accepting and might lead to some unexpected yet valuable answers.

Because codes based on sensemaking theories were utilised in the top-down approach, compared to the bottom-up approach, it was easier to discover themes that were rooted in theories. Therefore, its insights were more relevant to the research topic. It was also easier to generate themes in the top-down approach due to its preliminary codes. Nevertheless, the bottom-up approach was more enjoyable as it encouraged creativity in code generation. As more original insights were generated, unexpected inferences tend to emerge.

Several advantages of qualitative research could be identified through study. First of all, it is efficient in collecting data. For example, during a single short semi-structured interview, the direction of conversation could be consistently adjusted to explore intriguing experiences, which could result in valuable inferences. Another strength is its flexibility in conducting research. During the interview process, I was able to review and refine my interview style after 3 rounds of interviews. However, weaknesses were also spotted. Qualitative study is subject to bias if the sample quality is poor. Low generalisability was a major problem for this particular study due to the low diversity and small size of the sample. The analysis process is also labour-intensive. Due to the qualitative nature of the study, no analytical algorithm was used, hence the analysis was conducted manually.

Word count: 548/500

Coursework 2 – Part B: Quantitative HCI Research Methods

Section B1: Critically evaluating a study's method

Human operation errors on medical devices have been one of the underlying causes of preventable patient deaths every year (Kohn, Corrigan, & Donaldson, 2000), yet little attention was given to this field to address the pressing need of reducing unnecessary deaths. A study done by Wiseman et al., (2013) compared the usability of three number entry interfaces for medical infusion pumps and investigated if tailoring interfaces contribute to the increase of operational safety (i.e., reducing operational errors that could lead to tragic consequences). This essay aims to critically evaluate the study's methodology in terms of external and ecological validity.

The validity of a study refers to the approximate truth or usefulness of an inference (Trochim & Donnelly 2008). External validity measures the extent to which a study's inferences drawn from its sample apply to a broader population or other target populations (Findley et al., 2021). As a concept closely related to external validity, ecological validity examines whether the results of a study can be generalized to real-life settings (Lewkowicz, 2001). MacKenzie (2013) specified their difference by pointing out that "Ecological validity refers to the methodology, whereas external validity refers to the outcome". In this study about medical infusion pump interfaces, the target population are medical workers who are trained to operate the infusion pumps. However, 30 participants were recruited from the UCL Psychology subject pool as the sampled population. It is not expected the majority of them have the traits of a professional operator, for example, being trained to operate the pump under pressure.

Furthermore, most participants stated that they are familiar with the keypad and dial interface, but only 60% of participants stated that they are familiar with the chevron interface. While the chevron design is commonly encountered by professional operators (Masci et al., 2011), it is evident that the sampled population fell short of representing the target population.

Although it can be counterargued that, chevron interfaces are suggested to be inefficient in the professional working environment due to their low predictability (Masci et al., 2011), hence there could be little difference between non-professional and professional workers' performances on this type of interface, the flaw in the methodology still imposes a threat to the transportability of the lab results to the medical workers' performance in real-life settings. Therefore, a risk of low external validity exists. (Pearl & Bareinboim 2014)

During the experiment procedure, in order to recreate the condition in hospital wards where the speed and accuracy of number entries are both pursued, instructors choose to arbitrarily tell half of the participants to focus on speed and the other on accuracy. These artificially created conditions fall short to resemble the real-world situation as they are likely to create demand characteristic to the research. In hospital wards, professional careers always aim for both speed and accuracy while operating the infusion pump, as they understand these two metrics are equally critical to their job. They are trying their best to balance efficiency and accuracy, in order to achieve high performance. By splitting these two metrics apart, participants will tend to focus on elevating one metric of the performance in the dispense of the other, which is never the case in any hospital ward situation. In short, telling participants to only focus on one of the metrics that measures the overall user performance does not replicate the real-world pursuit of professional users, hence it would reduce ecological validity.

By extension, external validity is also affected by this experiment setting, as the generalisability of results is reduced by the failure of replicating the working environment of hospital wards in the study, namely the stress of professional workers and their skill to maintain equanimity. Participants are not professionally trained, nor are they put into a stressful environment, hence performance on these interfaces does not rigorously infer whether they will be preferred by real-world users. The consistently low error rates in the result might be the consequence of this flawed setting.

It is worth noting that for the materials used in the study, the numbers entered by participants were sampled from infusion pump logs in hospital wards. As ecological validity is described as the measurement of how much a method is generalisable to real-world settings, sampling data from a real-world dataset clearly adds ecological validity to this research.

While a few aspects of the methodology could negatively impact the ecological and external validity of this study, there are still numerous plausible settings that would enhance them. In the procedure, the numbers to be entered were presented in pairs as prescribed volume and rate, mimicking what is required on a real infusion pump. The interfaces manipulated were either common designs seen from infusion pumps or from other daily-life contexts. These two settings have significantly enhanced the ecological validity since they help establish a high relevance between the tested situation and the real-world situation.

Remarkably, although adopting a mixed design, while treating the within-subjects factor "tailoring", researchers utilised counterbalancing across the two conditions to cancel out any nuisance variables. Moreover, although the dataset was used repeatedly over two

conditions of the "tailoring" factor, they are randomised to reduce the carryover effects. Both measures focused on minimising the effect of participants unconsciously memorising some of the numbers presented and taking advantage in the next round. On one hand, these measures have increased the internal validity by eliminating possible confounding factors, on the other hand, they have increased the external validity by increasing the generalisability of the study results to a broad population of professional infusion device operators since numbers they encounter in real-world situations are in most case random, relying on memory to enhance performance hardly apply to any of these situations.

In conclusion, although the methodology of this study contains a considerable amount of design flaws that could damage its external and ecological validity, which might be due to the difficulty of replicating a reasonably complicated medical environment with limited human resources and apparatus, strong effort to build up the external and ecological validity are still observed. As is illustrated by Wiseman et al., (2013) "this study has been a preliminary step", the aim of this study is not to generate a robust interaction design solution to elevate the performance of infusion pump operators, but to show evidence of the possibility to do so and provide insights on how this could be done in the future. All in all, in terms of external and ecological validity, despite some noticeable imperfections, this is still a study of acceptable quality as there are numerous measures taken to enhance the two aspects of validity.

Word count: 1094/1000

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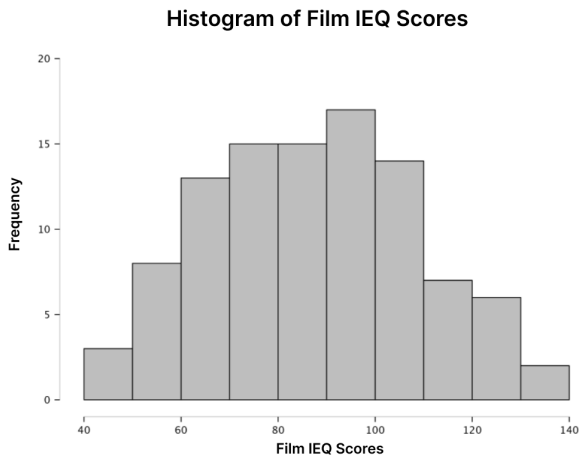
Section B2: Analysing a quantitative dataset

1. Summary statistics for all responses:
   a. Mean: 3.65
   b. Standard Deviation: 1.75
   c. Min: 1.00
   d. Max: 7.00

2. Film IEQ scores for each condition experienced by every participant:

   a. Film IEQ score when Participant is 1, Presentation is Interleaving and Background is Fun: 78
   b. Film IEQ score when Participant is 1, Presentation is Interleaving and Background is Pro: 82
   c. Film IEQ score when Participant is 2, Presentation is Floating and Background is Fun: 71
   d. Film IEQ score when Participant is 2, Presentation is Floating and Background is Pro: 109

3. Histogram based on each IEQ scores of each condition experienced by every participant:
   a. Histogram of Film IEQ Scores:



   b. Brief statement that describes the shape of the distribution:

      The shape of this histogram is very close to bell-shape, which indicates a normal distribution of the data.

4. Descriptive statistics for Film IEQ score for each condition:

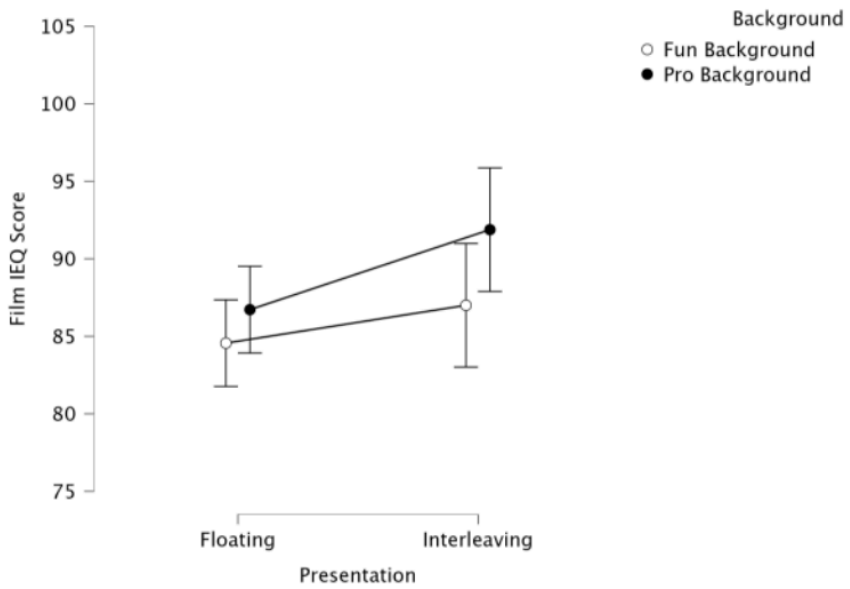
Descriptive Statistics for Film IEQ Scores

	Interleaving Presentation		Floating Presentation	
	Fun Background	Pro Background	Fun Background	Pro Background
Mean	87.0	91.9	84.6	86.7
Std. Deviation	22.5	23.6	19.9	20.6

5. A repeated measures ANOVA analysing the effect of Presentation and Background on Film IEQ scores suggests that :

    a. There was no significant main effect of Presentation on Film IEQ score, $F(1,48)=1.05$, $p=0.31$, which means the Film IEQ score is not effected by the type of presentation given.
    b. There was no significant main effect of Background on Film IEQ score, $F(1,48)=0.56$, $p=0.46$, which means the Film IEQ score is not effected by the type of background given.
    c. There was no significant interaction effect between Background and Presentation, $F(1,48)=0.16$, $p=0.69$, which means the type of background does not have a different effect on the Film IEQ score when the type of presentation changes, vice versa.

6. Graph that presents mean Film IEQ scores across Presentation and Background:



7. Descriptive statistics for comprehension scores for each condition:

Descriptive Statistics for Comprehension Scores

	Interleaving Presentation		Floating Presentation	
	Fun Background	Pro Background	Fun Background	Pro Background
Mean	18.00	18.20	17.4	17.1
Std. Deviation	3.69	3.99	2.5	2.3

Word count: 290/500

END